Autonomous Energy Researcher Agent Report

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Project Number: AAI - 13

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Academic Year: 2026

1. Problem Statement & Objectives

1.1 Problem Statement

In the modern digital era, the amount of information available on the internet is rapidly increasing, especially in domains like energy, sustainability, and environmental research. Extracting meaningful insights from this vast amount of unstructured data is a challenging and time-consuming task. Researchers and students often spend hours searching for reliable information, filtering irrelevant content, and summarizing findings manually.

This process not only reduces productivity but also increases the chances of missing important insights. Therefore, there is a need for an intelligent system that can automate the entire research workflow—from data collection to analysis and summarization.

1.2 Project Objectives

The main objective of this project is to build an intelligent and autonomous system that simplifies and accelerates the research process using modern AI technologies. By integrating a multi-agent architecture with web data collection tools and large language models, the system aims to handle complex research tasks efficiently. It focuses on transforming unstructured web data into structured and meaningful insights while minimizing manual effort. Additionally, the project emphasizes scalability, reusability, and accuracy, making it a practical solution for automated research in the energy domain and beyond.

The primary objectives of this project are:

- To design and develop an autonomous AI-based research system
- To implement a multi-agent architecture for efficient task handling
- To automate web data collection using search and scraping techniques
- To analyze and filter large volumes of unstructured data
- To generate structured and meaningful summaries using LLM
- To store research outputs in a knowledge repository for reuse
- To reduce human effort and improve research efficiency

1.3 Scope of the Project

The scope of this project is centered on developing an intelligent research system specifically for the energy domain, leveraging publicly available data sources. It is primarily designed for academic and research purposes, helping users quickly gather and understand relevant information. While the current implementation focuses on energy-related topics, the system is highly flexible and can be extended to other domains such as healthcare, finance, and

education. Additionally, the project serves as a practical demonstration of Agentic AI, showcasing how multiple AI agents can collaborate to solve complex real-world problems efficiently.

- The system focuses on energy-related research topics such as renewable energy, waste-to-energy, and sustainability
- It uses publicly available web data for research purposes
- The system is designed for academic, research, and learning use cases
- It can be extended to other domains such as healthcare, finance, and education
- The project demonstrates the practical implementation of Agentic AI systems

2. Proposed Solution

2.1 Key Features

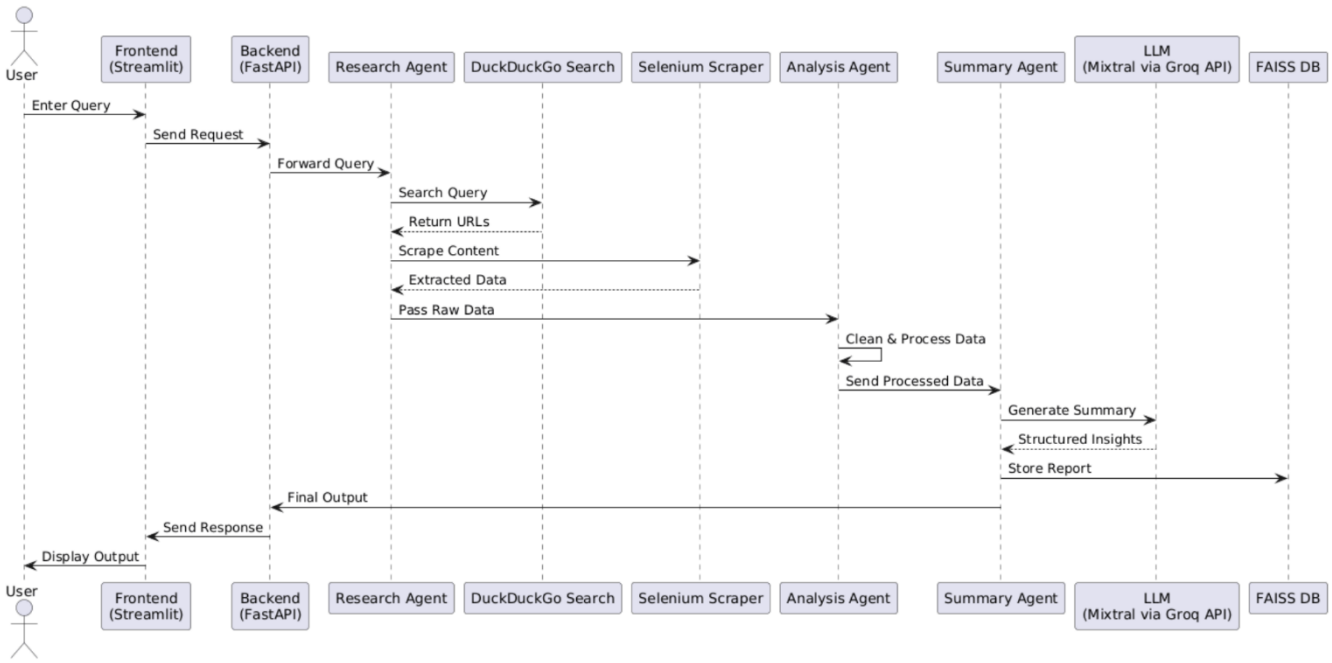
- Multi-agent system with defined roles
- Automated web search and scraping
- Intelligent data filtering and processing
- AI-powered summarization using LLM
- Knowledge storage using vector database (FAISS)
- Fast response generation using Groq API
- Modular and scalable architecture
- Reusable research outputs

2.2 Overall Architecture / Workflow

The system follows a layered and modular architecture to ensure scalability and maintainability.

Workflow Explanation:

1. The user enters a research query through the frontend interface
2. The frontend sends the request to the backend server
3. The backend forwards the query to the Research Agent
4. The Research Agent performs web search using DuckDuckGo
5. Relevant URLs are fetched and content is extracted using Selenium
6. The collected raw data is passed to the Analysis Agent
7. The Analysis Agent cleans, filters, and processes the data
8. The processed data is sent to the Summary Agent
9. The Summary Agent uses the LLM (Mixtral via Groq API) to generate structured insights
10. The final research report is stored in the FAISS knowledge repository
11. The output is returned to the frontend and displayed to the user



2.3 Tools & Technologies Used

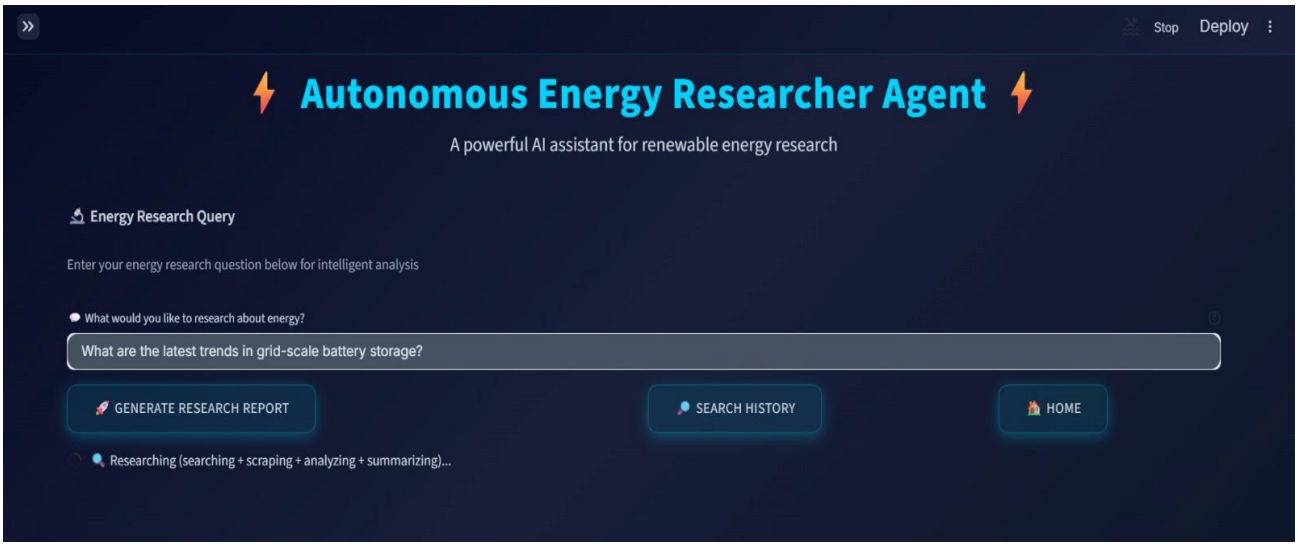
- Frontend: Streamlit
- Backend: FastAPI
- AI Framework: LangChain, CrewAI
- LLM: Mixtral model (via Groq API)
- Web Search: DuckDuckGo
- Web Scraping: Selenium
- Database: FAISS (Vector Database)
- Programming Language: Python

3. Results & Output

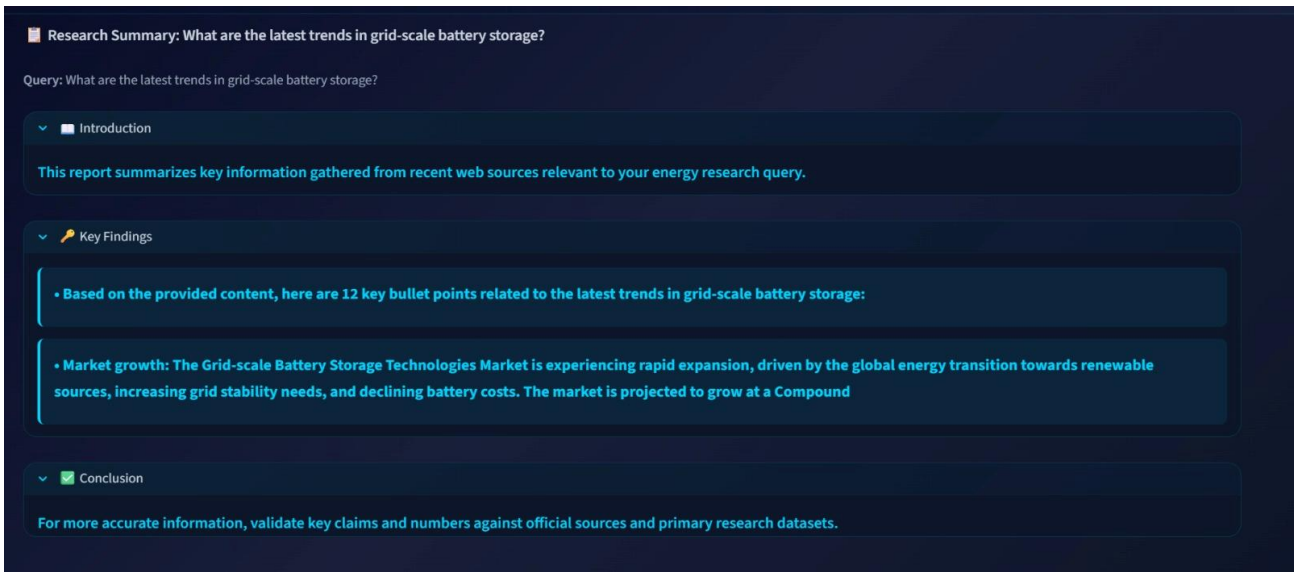
3.1 Screenshots / Outputs

- **Home Page Interface**
  Displays the main landing page introducing the system, its features, and example queries.

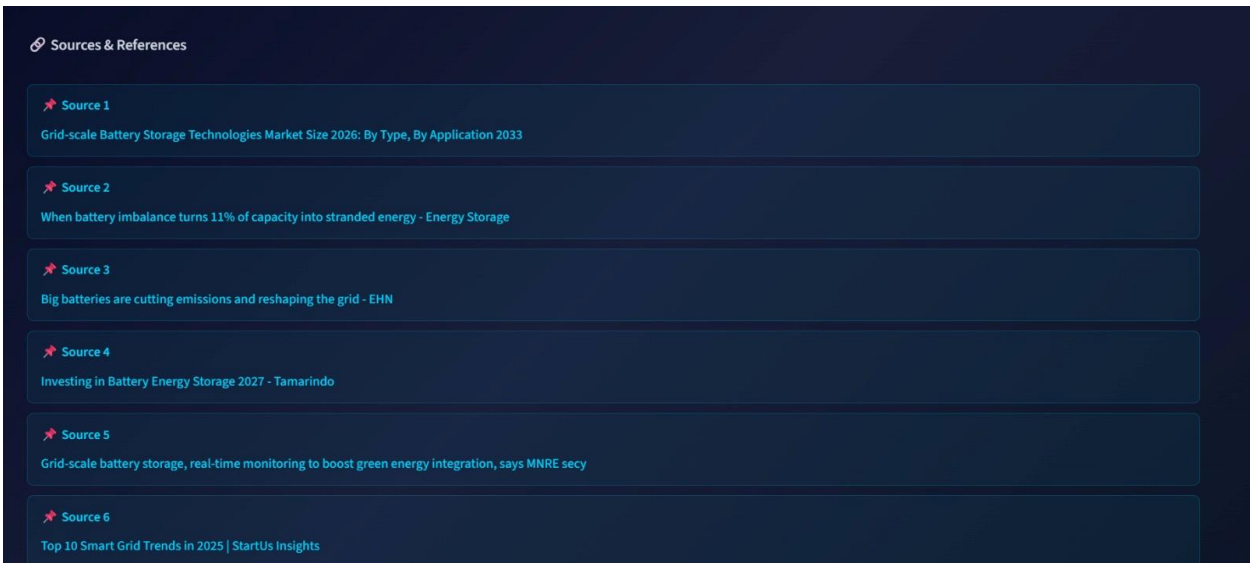
- **Processing / Execution State**
  Shows real-time system activity while performing search, scraping, analysis, and summarization.



- **Research Summary Output**
  Presents the final structured research report including introduction, key findings, and conclusion.

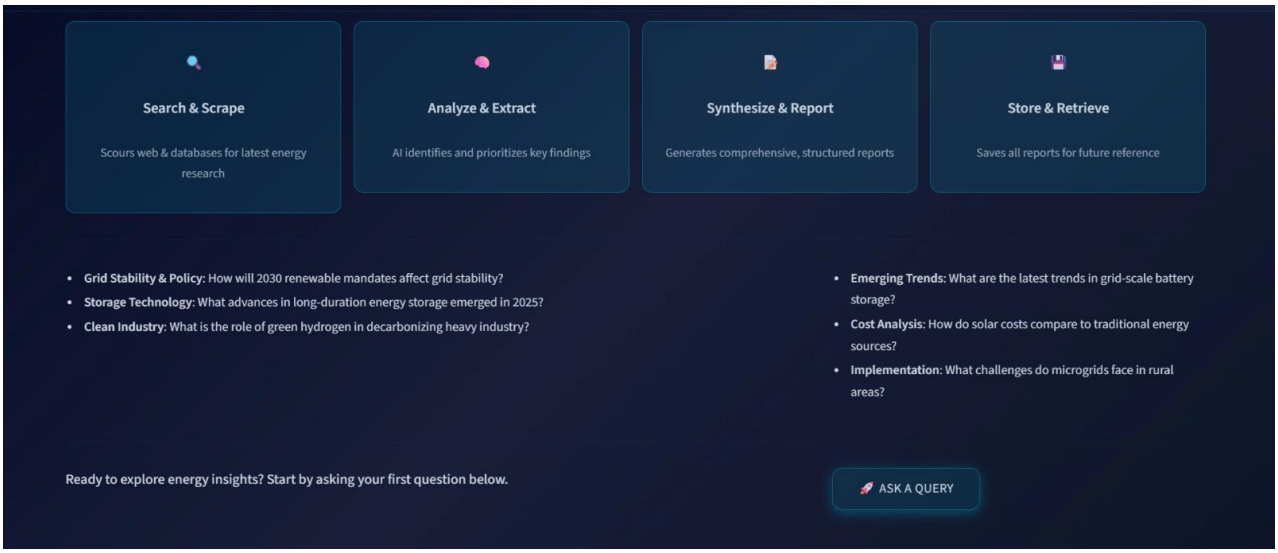


- **Sources & References Section**
  Provides a list of sources used to ensure transparency and credibility of the generated report.



3.2 Reports / Dashboards / Models

- AI-generated research summaries in structured format
- Extracted and processed web data
- Stored embeddings in FAISS database
- Logical flow of data processing through agents



3.3 Key Outcomes

- Significant reduction in manual research effort
- Faster data collection and analysis
- Improved accuracy in extracting relevant information
- Efficient storage and retrieval using vector database
- Demonstration of real-world application of Agentic AI

4. Conclusion

The Autonomous Energy Researcher Agent successfully demonstrates the integration of multiple AI components into a single intelligent system. By combining multi-agent

architecture, web data collection, and LLM-based summarization, the system automates the complete research workflow.

This project highlights the power of Agentic AI in solving real-world problems by improving efficiency, reducing manual effort, and enabling faster decision-making. It also provides a scalable foundation for building advanced AI-driven research systems in the future.

5. Future Scope & Enhancements

The project can be further improved with the following enhancements:

- Integration of more advanced and fine-tuned LLM models
- Addition of domain-specific agents for deeper analysis
- Implementation of real-time data streaming
- Development of advanced dashboards and visualizations
- Integration with external APIs for richer datasets
- Improvement in data accuracy and filtering mechanisms
- Deployment on cloud platforms for scalability
- Addition of user authentication and personalized research history
- Multi-language support for global usability